

The sustained causal bottleneck of in-context learning across modalities

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Abstract

Predicting whether a foundation model will exhibit strong in-context learning, and understanding its mechanism, remain open problems. Here we report that in-context learning in causal Transformers operates through a sustained, causally controlled one-dimensional channel of the residual stream — the *sustained causal bottleneck* (SCB) — whose formation is governed by the effective vocabulary entropy of inputs rather than by encoder/decoder architecture. Across sixty foundation models spanning text, protein, DNA and vision, low-entropy inputs produce a layer-spanning channel in protein and DNA encoders and in vision Transformers, while text encoder MLMs do not. Projection-out causally suppresses few-shot capability in Pythia, whereas spectral fine-tuning cannot induce a channel in BLOOM, an if-and-only-if signature. Spectral covariates predict eight benchmarks across forty-one language models with $R^2=0.86$. We release **scb-diagnostic**, deployable on any HuggingFace Transformer in under one minute.

1 Introduction

Transformer-based foundation models now span biomedicine: ESM2¹ predicts protein structure from amino-acid sequence, the Nucleotide Transformer² and Enformer³ annotate genomic regulatory elements, single-cell foundation models such as Geneformer⁴ and scGPT⁵ predict gene-network responses from sequencing data, vision Transformers⁶ drive medical-imaging benchmarks, and large causal language models increasingly serve as scientific assistants and clinical decision-support backbones^{7–11}. A capability that distinguishes these models from their pre-Transformer predecessors is *in-context learning* (ICL): solving a new task from a handful of demonstrations placed in the prompt, with no parameter update^{12–17}. ICL is what makes a single pre-trained model deployable across the long tail of biomedical tasks — rare-disease classification, low-resource sequence annotation, prompt-based retrieval — for which task-specific fine-tuning data is scarce or unavailable.

Yet the internal mechanism of ICL has remained opaque. Existing accounts attribute it to specific computational primitives — induction heads^{18,19}, function-vector heads^{20–23}, gradient-descent emulation^{24–26}, attention sinks²⁷, and training-dynamics phase transitions^{28–30} —

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and, almost without exception, study the text modality alone. Closely related concurrent work characterises a *Hidden Geometry* of ICL at the head/component level³¹. An equally fundamental but largely unanswered question is what representational *geometry* allows the relevant signals to propagate across dozens of transformer layers in the first place, and whether the same geometry exists in the protein, DNA and vision Transformers now standard in the wider biomedical community.

Here we provide an answer that is mechanistic, predictive, and modality-general: ICL in causal Transformers operates through a single *layer-spanning* one-dimensional channel of the residual stream — the *sustained causal bottleneck* (SCB) — and a single quantity, the effective vocabulary entropy V_{eff} , governs whether this channel forms. The SCB is to ICL what an axon is to a neural signal: a structural conduit through which the head- and component-level mechanisms of prior accounts must pass. We establish three properties that those head-level descriptions do not entail: eigenvector continuity (≥ 0.99) across the band, residual-stream-level projection-out causal control, and an if-and-only-if amplification result on a panel of 17 SCB-bearing causal LMs and 5 SCB-free causal-LM controls.

A separate strand of work has established that residual-stream covariance is heavy-tailed and dominated by “rogue” or “outlier” dimensions^{32–37}, and recent sparse-autoencoder interpretability extracts thousands of near-monosemantic features from the same residual stream^{38;39}. Neither line addresses (i) whether the dominant direction persists across many *consecutive* layers, (ii) whether the *same* direction is preserved across those layers, or (iii) whether the resulting structure has a load-bearing *functional* role; the SCB analysis below complements both by operating at the residual-stream-channel level.

We provide affirmative answers to all three using a series of nested probes on publicly released foundation models from protein, text, DNA and vision modalities (full panel composition in Methods §8.1). We define the **sustained causal bottleneck (SCB)** of a model as the maximal contiguous band of mid-to-late transformer layers for which the residual-stream covariance has participation ratio $\text{PR} \equiv (\sum \lambda_j)^2 / \sum \lambda_j^2 < 5$. The choice of 5 captures the regime in which the covariance is *quasi-rank-one*: at $\text{PR} = 1$ the covariance is exactly rank one; at $\text{PR} = 5$ a single direction still dominates the trace by approximately 60%.

We then ask three operational questions: **(Q1) Architecture.** Which model classes form an SCB? **(Q2) Geometry.** When an SCB exists, is it carried by the *same* direction across its layers? **(Q3) Function.** Is this direction load-bearing, and does its functional role depend on the prompting regime?

The answers organise into a five-part picture, summarised in Figure 1 and developed in Sections 2–6: SCBs form across modalities whenever the effective vocabulary entropy is sufficiently low, independently of the encoder/decoder dichotomy (Section 2); the SCB band carries one direction propagated through depth (Section 3); projecting it out cripples next-token prediction selectively (Section 4); the damage is amplified under few-shot relative to zero-shot prompting in proportion to SCB length (Section 5); and the channel and ICL ability emerge synchronously during pre-training (Section 6). Taken together the data support a single mechanistic claim: we identify a **sustained, causally-controlled** residual-stream channel that **scales with V_{eff} and generalises across modalities**, providing a structural

substrate for in-context learning. ICL is not a property of any particular head or layer but of a *channel*: a 1D subspace, sustained across many layers, through which task-specific signals are written and read.

Mechanistic intuition. The next-token gradient $\nabla_{x_L} \mathcal{L} = W^\top (\hat{p} - e_y)$ lives in the row-space of the unembedding W , which is at most V_{eff} -dimensional; iterated self-attention compounds rank-collapse⁴⁰ biased toward this row-space, so when one continuation class dominates at most positions the surviving variance concentrates onto a single principal component (Methods §8.7). The masked-LM objective lacks this temporal asymmetry, predicting (and we observe) $n_{\text{col}} = 0$ across all 16 encoder MLMs surveyed in Fig. 5b. Figure 1 summarises the four lines of evidence developed below.

Closely related prior accounts at the head, component, or token-outlier levels^{20–22;31;35} provide complementary descriptions but do not isolate a 1D channel sustained across many layers, demonstrate eigenvector continuity, or establish the if-and-only-if causal control reported here; we return to this comparison in the Discussion (Supplementary Table S16).

2 Effective vocabulary entropy, not the encoder/decoder dichotomy, governs SCB formation across modalities

Sustained one-dimensional channels form whenever the effective vocabulary entropy of a Transformer’s input distribution is low, irrespective of its training objective. The directional prediction of our V_{eff} -bottleneck theory (§7) is empirical, not architectural: a sufficiently low-rank token distribution drives any Transformer into the 1D-channel regime. We observe four classes: (i) *Causal LMs* on text or protein vocabularies ($V_{\text{eff}} \sim 10^4$): every one of the 17 deep-bootstrap panel forms an SCB (8–29 contiguous layers; Extended Data Fig. S2). (ii) *Text/protein encoder MLMs* (same high V_{eff} but no temporal asymmetry): 15/15 directly probed encoders return $n_{\text{col}} = 0$ (BERT-base/large, RoBERTa-base/large, DistilBERT, ALBERT, ELECTRA, DeBERTa, BioBERT, SciBERT, ESM2 8M/35M/150M/650M/3B). (iii) *DNA encoder MLMs* (low $V_{\text{eff}} \sim 4^6 = 4096$ kmers): both Nucleotide Transformer scales develop a strong SCB (NT-500M $n_{\text{col}} = 18$ across $L = 25$, 3-seed reproduced, $\text{PR}_{\text{max}} = 23$; NT-2.5B $n_{\text{col}} = 24$ across $L = 33$). (iv) *Vision Transformers* (low effective patch-token entropy): four of six probed vision models develop SCBs at $n_{\text{col}} \geq 3$ (ViT-base $n_{\text{col}} = 8$ across $L = 12$ on ImageNet; DINOv2-large $n_{\text{col}} = 6$, DINOv2-base $n_{\text{col}} = 3$, ConvNeXtV2-tiny $n_{\text{col}} = 4$). The encoder–decoder dichotomy is therefore not the discriminator: what predicts the SCB is the residual-stream covariance’s effective rank, which collapses under either temporal asymmetry (causal-LM training) or low-rank token distributions (DNA k-mer, image patches). Multilingual XLM-RoBERTa-base/large lie at threshold-noise ($n_{\text{col}} \in \{1, 3\}$), as expected for an intermediate V_{eff} .

To make the measurement reproducible, we use a uniform probe protocol (Methods §8.2): 200 modality-appropriate probe sequences (UniRef50, OpenWebText, nucleotide segments, ImageNet patches), top-100 stochastic eigendecomposition of the per-layer residual-stream covariance, and PR computation at every transformer layer. We define the SCB length n_{col} as the cardinality of the maximal contiguous block of mid-to-late layers (excluding the first 25% and last 5% of depth) with $\text{PR} < 5$. Threshold sensitivity is reported in Supplementary §S9:

tightening to $PR < 3$ preserves all qualitative claims (Spearman $\rho \geq 0.97$ across model rankings).

Text and protein encoder MLMs do not form an SCB. Of the 17 text/protein encoder MLMs we probed (BERT-base/large, RoBERTa-base/large, DistilBERT, ALBERT-base, ELECTRA-base, DeBERTa-v3-base, BioBERT, SciBERT, ESM2 8M/35M/150M/650M/3B, plus the two XLM-RoBERTa scales), 15 return $n_{\text{col}} = 0$ with $PR > 30$ throughout depth (PR_{max} 35–68). The two XLM-RoBERTa runs — base ($n_{\text{col}} = 3$, PR_{max} 44.7) and large ($n_{\text{col}} = 1$, PR_{max} 60.4) — sit at the threshold-noise boundary: the multilingual vocabulary adds a faint language-identifier direction but no sustained band-spanning channel.

DNA encoder MLMs form an SCB under the V_{eff} prediction. Both Nucleotide Transformer scales we probed develop a strong SCB. NT-500M reproducibly returns $n_{\text{col}} = 18$ across $L=25$ in three independent seeds (PR_{max} identical to two decimals at 23.0, 23.0, 23.0), and NT-2.5B returns $n_{\text{col}} = 24$ across $L=33$ (PR_{max} 23.2). The DNA 6-mer tokenizer collapses the residual stream onto $V_{\text{eff}} \sim 4^6 = 4096$ kmer states, two orders of magnitude smaller than the typical text MLM ($V_{\text{eff}} \sim 3 \times 10^4$); the gradient-rank-bound theory of §7 predicts that any sequential model trained on so few effective token classes will collapse to a 1D channel regardless of objective. The Nucleotide Transformer is therefore not a contradiction of the theory but a positive confirmation: the SCB tracks low effective vocabulary entropy, not encoder–decoder architecture. Concurrent behavioural work⁴¹ demonstrates that genomic next-token predictors exhibit ICL; we identify the residual-stream mechanism (low V_{eff} via 6-mer tokenizer) underlying this empirical finding.

Causal LMs split three ways. The SCB length distribution across the deep-bootstrap causal-LM panel (Extended Data Fig. S2) is trimodal. *Heavy SCB* (n_{col} spanning $> 80\%$ of mid-to-late layers): Mistral-7B-v0.3 (29 collapsed layers), Pythia-2.8B (28), Pythia-6.9B (28), Phi-1.5 (21). *Medium SCB*: Pythia-410M through Pythia-1.4B, OLMo-7B, TinyLlama-1.1B, SmolLM-1.7B (open-decoder families building on the Llama-2⁴² recipe). *Minimal or no SCB*: Pythia-70M/160M, BLOOM-1B1, Qwen-2.5-0.5B/1.5B, GPT-Neo-2.7B, StableLM-2-1.6B; the five SCB-free among them form the IFF control cluster (§5).

MoE forms SCB; Mamba shows borderline behaviour. Both MoE models tested (Qwen-MoE-A2.7B, OLMoE-1B-7B) form an SCB, with Qwen1.5-MoE in particular showing $R = 3.02$, *higher* than the dense Mistral-7B $R = 2.54$. This is despite MoE’s structurally different per-layer computation (sparse expert routing rather than dense MLP). The implication is that SCB is sustained at the level of the residual stream, not at the level of per-layer inner computation, and that experts in MoE *coordinate* their additive contributions to that residual stream onto a shared 1D direction. Mamba+attention hybrid (Zamba2-1.2B) shows borderline SCB ($n_{\text{col}} = 1$, $R = 0.62$), similar to BLOOM-1B1 / GPT-Neo-1.3B border cases.

3 The SCB is one direction, propagated through depth

The leading eigenvector at one collapsed layer points in nearly the same direction as the leading eigenvector at every other in-band layer (within-band $\langle |\cos(v_\ell, v_{\ell'})| \rangle = 0.993$ for Mistral-7B-v0.3, our most heavily probed exemplar). The same measurement on encoder MLMs returns 0.332 for ESM2-3B and similar values for the other encoders (Fig. 2c)

— indistinguishable from the random-orientation expectation $\sqrt{2/(\pi D)} \approx 0.01$ for hidden dimension $D \approx 4096$. The contrast establishes the SCB as a single 1D channel propagated through depth, not a stack of layer-local rank-one accidents.

For each model we computed the leading eigenvector at every collapsed layer (deterministic SVD seed; Methods §8.3) and its cosine similarity to the leading eigenvector at every other collapsed layer.

Causal-LM exemplars. Mistral-7B-v0.3 ($L = 33$, $n = 3$ seeds in the NxN panel) attains a within-band mean cosine of 0.993 across the SCB band (layers 7–29; Fig. 2a). Pythia-1B ($L = 17$, 151 consec-cosine seeds) reaches 0.95 ± 0.04 in its $\ell \in \{4, \dots, 14\}$ band; OLMo-2-7B and Phi-1.5 reach 0.91 and 0.96 under the same protocol. The four largest panel causal LMs (Mistral-7B-v0.3, Pythia-6.9B, Pythia-12B, ProGen2-xlarge) all exceed 0.99 within-band, and no SCB-bearing model drops below 0.82. Cross-band cosines (between an SCB layer and a non-SCB layer in the same model) collapse to 0.4 ± 0.2 , confirming that the channel is contained inside the band rather than smeared across all layers.

Encoder-MLM contrast. ESM2-3B ($L = 37$) returns a within-band mean cosine of 0.332; BERT-large ($L = 24$) returns a comparable 0.33 ± 0.09 . Both sit within numerical noise of the random-orientation expectation ($\sqrt{2/(\pi D)} \approx 0.01$ for $D = 1024$). The 0.99 versus 0.33 separation across the 16-model cosine-continuity panel (12 causal + 4 encoder; Fig. 2c) is the single strongest piece of evidence that the SCB is a propagated channel rather than a sequence of independent layer-local accidents.

4 The 1D channel is load-bearing

Projecting the SCB direction out at every collapsed layer raises next-token NLL by 49–702% across 19 causal LMs, a damage two to three orders of magnitude larger than that caused by ablating any other direction. The same projection-out applied to the second principal direction or to a random orthogonal direction is essentially benign (median $7.8\times$ and $480\times$ smaller, respectively), confirming that the SCB is the residual stream’s main load-bearing channel rather than one of many.

We apply a *projection-out* hook (Methods §8.5) at every SCB layer of each causal LM and measure the resulting next-token NLL on a held-out 200-token continuation corpus.

Leading direction is uniquely load-bearing. Across the 19 SCB-bearing causal LMs ablation-tested, leading-eigvec projection-out raises next-token NLL by 49–702%, while the second eigenvector and a random orthogonal direction are $7.8\times$ and $480\times$ less damaging (medians). Pythia scales monotonically from 49% (70 M) to 702% (12 B); Pythia-12B at 30 collapsed layers is the most-damaged model in the panel.

4.1 Asymmetric fine-tune control: suppress works, induce fails

Spectral-suppress fine-tuning on Pythia-1B ($\mathcal{L}_{\text{LM}} + \alpha \lambda_1 / \sum_k \lambda_k$, $\alpha = 1.0$, 3 000 steps; cf. ⁴³) collapses $n_{\text{col}}: 11 \rightarrow 0$ on 6/6 seeds with λ_1/λ_2 dropping $683 \rightarrow 2.4$. The converse spectral-induce on BLOOM-1B1 leaves $n_{\text{col}} = 15$ unchanged on 10/10 seeds at $\alpha \in \{0.1, 1.0\}$, even though the raw λ_1/λ_2 amplitude moves dose-dependently: the induce loss adds spectral mass within the existing band rather than nucleating a new layer-spanning direction. The control is therefore *asymmetric*: we can remove a trained-in SCB but cannot retrofit one onto a model that

did not develop it during pre-training, supporting the structural-consequence picture of §S6. Per-seed values, dose-response curves and the full fine-tune panel are in Supplementary §S7 and Extended Data Fig. S4.

5 SCB length predicts the few-shot/zero-shot amplification ratio

The damage caused by projecting the SCB out is $2\text{--}7\times$ larger under few-shot prompting than under zero-shot prompting, and the magnitude of this amplification scales linearly with SCB length across $n = 17$ causal LMs (Pearson $r = 0.836$, $p = 2.9 \times 10^{-5}$; deep-bootstrap panel, 5 829 source records). Five SCB-free causal LMs (BLOOM-1B1, Qwen2.5-0.5B, Qwen2.5-1.5B, GPT-Neo-2.7B, OLMo-7B) all sit at $R \approx 1$ on the baseline few-shot/zero-shot loss ratio (no projection — $\Delta\text{NLL}_{\text{ZS}}=0$ makes projection-out R mathematically undefined for $n_{\text{col}}=0$), establishing an if-and-only-if causal control.

For each causal LM we computed two NLLs: (i) a zero-shot NLL on a 200-token English continuation task; (ii) a few-shot NLL on the same continuation task preceded by 3 exemplar input/output pairs. ΔNLL is the increase under leading-eigenvector projection-out; the amplification ratio is $R \equiv \Delta\text{NLL}_{\text{ICL}}/\Delta\text{NLL}_{\text{ZS}}$.

The headline observational result: across the paper-canonical 17-model panel (deep bootstrap, $n_{\text{seeds}} \geq 10$ per cell, 5 829 source ablation records), Pearson $r(n_{\text{col}}, R) = 0.836$, $p = 2.87 \times 10^{-5}$, $n = 17$, with bootstrap 95% CI [0.68, 0.93] over 5 000 resamples (Extended Data Fig. S2). On the broader 23-model sample (mixed n_{seeds} from cycle-17 era) the same correlation gives $r = 0.59$, $p = 8 \times 10^{-4}$, $n = 23$, preserved across PR thresholds 3/5/10 (Supplementary §S1).

The five $n_{\text{col}} = 0$ models (BLOOM-1B1, Qwen2.5-0.5B, Qwen2.5-1.5B, GPT-Neo-2.7B, OLMo-7B) provide a powerful negative control: their baseline few-shot/zero-shot loss ratio is 1.00 across all five (the projection-out R is mathematically undefined because $\Delta\text{NLL}_{\text{ZS}} = 0$ when there is no SCB to project out). Conversely, every SCB-forming model with $n_{\text{col}} \geq 8$ shows $R \geq 1.49$. Together this is an *if-and-only-if* structural claim:

$$n_{\text{col}} > 0 \iff \text{ICL amplification by leading-eigvec projection-out exists.} \quad (5.1)$$

Family-level breakdown and specificity controls. Within Pythia (8 scales) R rises monotonically with n_{col} ($r > 0.85$); Mistral-7B-v0.3 gives the second-largest panel amplification ($R = 2.54$ at $n_{\text{col}} = 29$); BLOOM spans the IFF-control left edge ($n_{\text{col}} \in \{0, 1, 2, 3\}$, $R \in [0.78, 1.54]$); the remaining panel cuts diagonally across the same regression line. Projecting out the second, fifth, or a random orthogonal direction at the same layers gives $R = 1.13 \pm 0.27$, 0.94 ± 0.31 , 1.02 ± 0.35 respectively (full per-family breakdown and specificity panel in Extended Data Fig. S2d).

6 The channel and ICL ability emerge synchronously during pre-training

Across six Pythia pre-training trajectories (70 M to 6.9 B), the SCB direction crystallises abruptly between training steps 2 000 and 4 000, the same window in which few-shot ICL amplification first appears, and then grows in magnitude over the next 5×10^4 steps without changing direction (cosine 0.997 between step-3 000 and step-143 000 leading eigenvectors). The trajectory is two-phase: an early *direction lock* that establishes the channel, followed by an *amplitude growth* phase in which task-relevant signal is loaded into the now-fixed channel.

We probed six Pythia⁴⁴ scales (70 M, 160 M, 410 M, 1 B, 1.4 B, 6.9 B) at 30+ pre-training checkpoints each (Methods §8.9; full panel in Extended Data Fig. S3). The trajectory is two-phase: *Phase I* (step 2 000→4 000) jumps n_{col} from 0 to 8 with direction-lock cosine 0.997 between step-3 000 and step-143 000 leading eigvecs; *Phase II* (step 4 000→30 000) stabilises n_{col} at 11–12 while λ_1/λ_2 grows from 8 to 421 and R climbs $1.25 \rightarrow 2.43$. Mature R across the six scales is $\{1.07, 1.21, 1.62, 2.04, 2.31, 2.70\}$ (Pearson $r(\log N, R) = 0.92$, $n = 7$ including Pythia-12B); the $A_{\text{final}}-R$ correlation is $r = 0.94$, so SCB amplitude growth accounts for $\sim 88\%$ of the cross-scale variation in mature R .

6.1 Application A: SCB onset as early-warning ICL probe

The geometric quantity n_{col} measured at small scale predicts the mature R of larger models in the same family. Across our 16-model panel (Extended Data Fig. S5), a leave-one-out regression of mature R on n_{col} alone gives $R^2 = 0.62$ ($r = 0.79$, $p = 2.8 \times 10^{-4}$), versus $R^2 = -0.02$ for $\log N$ alone^{45;46}. The same observation holds in time: across six Pythia trajectories the lead factor (ICL-maturity step / SCB-onset step) has median $1.10\times$ but reaches $6\times$ at Pythia-6.9B and $15\times$ at Pythia-1.4B (Extended Data Fig. S6). A single residual-stream eigendecomposition is two to three orders of magnitude cheaper than a full ICL benchmark, so even at $1\times$ lead the SCB metric is a 10^2 – 10^3 cheaper proxy.

7 Discussion

The SCB is the residual stream’s main carrier. Mistral-7B’s residual stream nominally has $D = 4096$ dimensions, yet a single direction captures 99.998% of its in-band variance and bears the bulk of the ICL load. The 4096-dimensional residual stream computes through what is, in this regime, a one-dimensional channel.

Why one direction. The 1D limit follows from a four-step gradient-rank argument (Methods §8.7, Supplementary §S6). The next-token loss gradient $\nabla_{x_L} \mathcal{L} = W^\top (\hat{p} - \mathbf{e}_y)$ lives in the row-space of the unembedding W , which bounds the residual-stream covariance rank by the effective vocabulary entropy:

$$\text{rank}(\text{Cov}(x_L)) \leq V_{\text{eff}}. \quad (7.1)$$

Once the autoregressive objective concentrates this surviving rank onto a single continuation-direction principal component, the V_{eff} -bottleneck theory predicts that mature n_{col} saturates

at the same fraction of model depth attained by mature pre-trained causal LMs at this scale,

$$n_{\text{col}}^{\text{sat}} \approx L \cdot 0.6, \quad (7.2)$$

empirically anchored by the saturation-amplitude fit $r_{\text{sat}} \approx A(1 - e^{-(N/N_0)^\alpha})$ across 8 Pythia scales ($R^2=0.83$; Fig. 5). The same argument predicts that low effective vocabulary entropy alone suffices to drive a 1D channel, with or without temporal asymmetry. The DNA-encoder and ViT positives reported in §2 test this prediction; XLM-RoBERTa’s threshold-noise behaviour and the encoder-MLM null are likewise refinements of the same V_{eff} test (full discussion in Supplementary §S6).

Relation to prior work. Head-, component-, token-outlier and feature-level accounts^{18;22;31;35–39;47–52} provide complementary descriptions; none isolate the across-many-layers 1D channel, demonstrate eigenvector continuity, or establish the if-and-only-if causal control reported here. Concurrent work has identified low-dimensional ICL subspaces in specific tasks^{53;54} and along generation trajectories^{55;56}; we extend these findings by showing the channel is sustained across mid-to-late layers, causally controllable, and predicted by V_{eff} across modalities. A point-by-point differentiation appears in Supplementary Table S16.

Limitations. (a) A few cells of the broader 23-model panel rest on 1–3 seeds; all results carry bootstrap CIs. (b) Few-shot prompts use a continuation task; generalisation to translation, classification, and arithmetic ICL prompts is an open follow-up. (c) Spectral-induce fails to retrofit a true SCB (Supplementary §S7). (d) Application-A’s early-warning lead is scale-dependent. (e) The first-principles theory is heuristic (Methods §8.7); a closed-form derivation is open. Probe-corpus and probe-protocol robustness checks are in Methods §8.6 and Supplementary §S5.

Outlook. Viewing ICL as a 1D-channel property — not a head- or circuit-property — reframes several mechanistic-interpretability questions. The channel furnishes prior head-level findings (induction heads, function-vector heads, attention sinks, massive activations) with a common substrate to write into. It also supplies a single summary statistic, n_{col} , that predicts ICL amplification more reliably than parameter count alone.

Three immediate consequences follow. *First*, the V_{eff} -bottleneck theory makes a testable saturation prediction at $n_{\text{col}} \approx L \cdot 0.6$ which we test in Application B below. *Second*, the same probe transfers to new architectures and modalities without modification, so n_{col} serves as an early-warning ICL monitor (Application A; Extended Data Fig. S6); a 1-minute diagnostic CLI is released as Application C. *Third*, the asymmetric causal control (suppress works, induce fails) supports the structural-consequence reading of the autoregressive objective (Supplementary §S12).

Application B: From-scratch test of the V_{eff} saturation prediction

To probe the V_{eff} relationship causally we ran an equal-architecture, equal-compute from-scratch pretraining sweep on a 160-M-parameter Pythia-style decoder ($L = 12$), holding data (FineWeb-edu, 5 B tokens), optimiser (AdamW, lr 6×10^{-4} , cosine), and architecture fixed, and varying only the tokenizer vocabulary across four orders of magnitude: $|V| \in \{256, 1\,024, 8\,192, 50\,257, 100\,000, 200\,000\}$, 4–8 seeds per cell, 35 runs total (Methods §8.12,

Supplementary §S12). The pre-registered prediction was that mature n_{col} should saturate at $L \cdot 0.6 \approx 7$ in the regime where 5-B-token pretraining reaches mature ICL ($R \geq 1.5$). The data confirm the prediction across the entire mature-ICL band: the four cells from $|V|=8\,192$ to $|V|=200\,000$ all sit at $\bar{n}_{\text{col}} \in [6.57, 7.12]$ with $\bar{R} \in [2.49, 2.73]$ (all 27 mature seeds clear $R \geq 1.5$); the $|V|=1\,024$ cell sits on the rising edge ($\bar{R}=1.65$); the $|V|=256$ byte-level cell does not reach mature ICL ($\bar{R}=1.10$, $n_{\text{col}}=0$ on 4/4 seeds).

Application C: 1-minute SCB diagnostic CLI

We release `scb-diagnostic`, a minimal CLI tool that takes any Hugging Face Transformer identifier and returns $(L, n_{\text{col}}, \hat{R}, V_{\text{eff}} \text{ regime})$ in under a minute (200-probe scan + per-layer PR). $\hat{R}$ comes from the leave-one-out regression of the 17-model deep-bootstrap panel: $\hat{R} = 0.69 + 0.064 \times n_{\text{col}}$ (LOO $R^2 = 0.55$). The tool is intended as a cheap proxy for the full ICL benchmark sweep, particularly during the early-warning window (step 2 000–4 000 of large-model pretraining). Source repository <https://github.com/Jie-Ni/biolm-scb>; the entire diagnostic runs in < 5 GB GPU memory and on CPU as well.

Application D: Biomedical few-shot QA

The V_{eff} -bottleneck account predicts that in-domain biomedical pretraining *reduces* rather than enhances the SCB saturation amplitude. A four-model panel on PubMedQA and MedQA-4opt (general-purpose Pythia-1B/2.8B vs. BioGPT-Large¹⁰ and BioMedLM¹¹) confirms this prediction: the general-purpose Pythia-2.8B outperforms both biomedical-specialised models on PubMedQA (0.620 vs. 0.480 and 0.418). Full evaluation table, methodology and provenance in Methods §8.13 and Supplementary §S13.

8 Methods

8.1 Panel composition

We use four nested panels of publicly released foundation models, each serving a different claim. Panel sizes differ because the measurements differ in cost: a single-layer participation-ratio scan is cheap (one forward pass), an eigenvector-continuity NxN cosine matrix requires multiple seeds of dense rank-100 SVD per layer, an IFF projection-out ablation requires few-shot/zero-shot probe pairs under the leading-direction hook for every model, and the trajectory panel requires re-loading every Pythia checkpoint at every step. All panels are nested subsets of the same source model inventory.

Each panel reports the per-model count actually probed by its underlying measurement; numbers below were re-derived from the source-data files listed in `source_data/`.

Participation-ratio architectural scan (Hero Fig. 1b background, Extended Data Fig. ED1, Fig. 5b): **60 models with full per-layer PR profiles**, aggregated into the file `fig123_alllayer_v3_full.npz` from three sources — (i) 16 cycle-32 baseline models (file `fig123_alllayer.npz`); (ii) 20 fresh probes added in cycle-74 on LEO5 via `_leo5_pr_probe.py` (3 seeds for the Nucleotide Transformer reproducibility check); and (iii) 49 models from the MUSICA-INN `vit_alllayer/`, `progen_alllayer/`, `alllayer_round2/` probe runs (3–5 seeds per model; modality-matched probe corpora: OpenWebText for text models, UniRef50 for protein, ImageNet for vision, k -mer DNA for Nucleotide Transformer).

Modality split: 35 text models, 6 protein, 6 vision, 2 DNA; architecture split: 27 decoder, 33 encoder. $n_{\text{col}} > 0$ holds across all four input regimes the V_{eff} theory predicts: 17/17 causal LMs (text + protein), 2/2 DNA encoder MLMs (NT-500M $n_{\text{col}} = 18$, NT-2.5B $n_{\text{col}} = 24$), and 4/6 vision encoders (ViT-base $n_{\text{col}} = 8$, DINOv2-large $n_{\text{col}} = 6$, DINOv2-base $n_{\text{col}} = 3$, ConvNeXtV2-tiny $n_{\text{col}} = 4$). $n_{\text{col}} = 0$ holds for the high- V_{eff} encoder class as predicted: 15/15 text/protein encoder MLMs (BERT-base/large, RoBERTa-base/large, DistilBERT, ALBERT-base, ELECTRA, DeBERTa-v3, BioBERT, SciBERT, ESM2 8M/35M/150M/650M/3B). XLM-RoBERTa-base/large ($n_{\text{col}} \in \{1, 3\}$) and DINOv2-small/Swin-base ($n_{\text{col}} \in \{0, 2\}$) lie at threshold-noise.

Consecutive-cosine panel (Fig. 2c-d cross-model backgrounds, Supplementary §S2): **41 models in leo5_consec_cos.npz**: 23 causal LMs spanning Pythia, BLOOM, Qwen, GPT-Neo, Mamba, RWKV, OPT, SmolLM, TinyLlama, Phi, StableLM, plus 18 encoder MLMs (ESM2 5-scale, BERT-base/large, RoBERTa-base/large, XLM-RoBERTa, BioBERT, BiomedBERT, ClinicalBERT, SciBERT, ALBERT-base/large, DistilBERT, ELECTRA, DeBERTa). Used for cross-model trends only — the metric is consecutive $|\cos(v_\ell, v_{\ell+1})|$, which does not distinguish causal from encoder cleanly because consecutive layer eigenvectors drift slowly in both architectures.

Cosine-continuity NxN panel (Fig. 2a,b): **16 models in fig2_nxn.npz** with ≥ 1 seed of dense rank-100 SVD per layer for the all-pair cosine matrix: 12 causal LMs (Pythia-70M/160M/410M/1B/1.4B/6.9B, Mistral-7B-v0.3, BLOOM-3B/7B1, GPT-Neo-2.7B, TinyLlama-1.1B, Phi-1.5, Qwen-2.5-7B) and 4 encoder MLMs (BERT-large, RoBERTa-large, ESM2-650M, ESM2-3B). In-band off-diagonal $\langle |\cos(v_\ell, v_{\ell'})| \rangle = 0.993$ for Mistral-7B and 0.332 for ESM2-3B (the two exemplars in Fig. 2a,b); the cross-model bar in Fig. 2c reports per-model values for all 16.

Phase-2 ablation panel (Fig. 3): **42 model-tags in fig3_phase2.npz** have valid **leading-direction ΔNLL data**; the bar chart in Fig. 3b shows 10 representative SCB-bearing causal LMs covering the Pythia 8-scale series, Mistral, SmolLM-1.7B, StableLM-2-1.6B and TinyLlama-1.1B for compactness. The full panel of 42 is reported in source-data CSV.

IFF deep-bootstrap panel (Extended Data Fig. S2): $n_{\text{IFF}} = 17$ **SCB-bearing** causal LMs with $n_{\text{seeds}} \geq 10$ per leading-direction projection-out cell (8 Pythia + 3 BLOOM + Mistral-7B-v0.3 + StableLM + TinyLlama + SmolLM-1.7B + Phi-1.5 + Phi-1). Plus **5 SCB-free IFF controls** ($n_{\text{col}} = 0$, baseline metric): BLOOM-1B1, Qwen-2.5-0.5B, Qwen-2.5-1.5B, GPT-Neo-2.7B, OLMo-7B. Total source ablation records across both groups: 5 829.

Trajectory panel (Extended Data Fig. S3): seven Pythia scales (70 M, 160 M, 410 M, 1 B, 1.4 B, 2.8 B, 6.9 B), each at 30+ pre-training checkpoints. Pythia-12B is included only at the final checkpoint by compute budget.

Cross-panel coverage. The union of probed models across all panels is 41 causal LMs (overlapping with 18 encoder MLMs from the consec-cosine panel and 4 encoders from the PR-profile scan; total $41 + 18 = 59$ unique models with $\geq$ one of the substrate metrics). Per-figure source-data CSV/JSON files in **source_data/** list every model in the corresponding panel together with raw n_{col} , λ_1/λ_2 , R , and per-seed values for reproducibility.

8.2 Probe corpus and preprocessing

For each model, we use a fixed probe corpus to estimate the residual-stream covariance. The corpus is modality-matched and held constant across all probe runs to ensure that any change in n_{col} , leading eigenvector direction, or participation ratio is attributable to the model’s training-induced residual-stream geometry rather than to probe-corpus drift. **Text causal LM and encoder MLM**: 200 sequences of 64–128 tokens drawn from OpenWebText⁵⁷ with the model’s native tokenizer; sequences are truncated at 128 tokens and optionally padded to a uniform length using the model’s pad token (or eos token when no pad token is defined). **Protein causal LM** (ProGen2 family): 200 UniRef50 fragments of 50–200 amino acids, tokenized with the model’s native tokenizer (typically character- or BPE-on-amino-acid alphabet). **Protein encoder MLM** (ESM2): same UniRef50 fragments. **DNA encoder MLM** (Nucleotide Transformer): 200 nucleotide sequences of 200–1 000 bp using the model’s k -mer tokenizer. **Vision Transformer** (ViT, DinoV2, Swin): 200 ImageNet-1k validation patches processed via the model’s native preprocessing pipeline. Probe-corpus invariance is verified in Supplementary §S5: the rank-ordering of n_{col} across 28 causal LMs under three text-domain corpus alternatives (English, multilingual, code) preserves Spearman $\rho \geq 0.99$ and the IFF set is unchanged.

For each (model, sequence) pair we forward-pass the input and extract per-layer hidden states `h_layer[batch, token, dim]` from `model.transformer.h` or the model-specific equivalent (per-architecture mapping in §8.8). Hidden states are collected at all token positions across all 200 sequences as samples in an (N, D) matrix (e.g. $N \approx 200 \times \text{seq_len} \approx 12\,800$ for text and $\approx 30\,000$ for protein). Hidden states are stored in float32 to avoid SVD numerical instability and minimise dynamic-range loss for the trailing eigenvalues.

8.3 Stochastic eigendecomposition

For each layer’s (N, D) hidden-state matrix H , we centre $H_c = H - \bar{H}$ and compute the leading 100 eigenvalues and right-singular vectors of the covariance $\Sigma = H_c^\top H_c / (N - 1)$ via stochastic truncated SVD implemented in `torch.svd_lowrank` with parameters $q = 100$, $niter = 4$, and a deterministic per-(model, layer, seed) random number generator. We use `torch.linalg.svdvals` as a fallback for the rare layer-seed combinations where stochastic SVD fails to converge under bf16 input casting (re-cast to fp32 then retry). Eigenvalues are reported in descending order; eigenvectors are stored as right-singular vectors of $H_c / \sqrt{N - 1}$. The choice of top-100 is justified by the empirical observation that $\lambda_1 / \lambda_{100}$ typically exceeds 10^4 for SCB-bearing models, so contributions beyond the top-100 are below floating-point numerical precision for the participation-ratio statistic.

For the trajectory analyses in Section 6 and Figure S3, the same protocol is applied at every pre-training checkpoint, with the `revision="step{N}"` keyword passed to `from_pretrained` (Pythia checkpoint convention; §8.9).

8.4 SCB definition and PR sensitivity

The participation ratio at layer ℓ is

$$\text{PR}_\ell = \frac{(\sum_k \lambda_{\ell,k})^2}{\sum_k \lambda_{\ell,k}^2}, \quad (8.1)$$

which equals 1 when the covariance is exactly rank-1 (a single direction) and equals D in the maximal-spread isotropic case. The empirical observation motivating SCB is that $\text{PR}_\ell < 5$ is reached at *contiguous* mid-to-late layers in causal Transformers but is essentially never reached in encoder MLMs.

We define the **SCB length** n_{col} as the cardinality of the maximal contiguous block of layers $\ell \in [[0.25 L], [0.95 L]]$ for which $\text{PR}_\ell < 5$, where L is the model’s total transformer layer count. The mid-to-late window (excluding the embedding-near first 25% and the final 5%) avoids boundary effects: the embedding layer is dominated by tokenization geometry, and the final layer is dominated by the unembedding pre-softmax projection. PR threshold sensitivity is reported in Supplementary §S1 (PR < 3, 5, 10); rank-orderings across all 28 causal LMs are stable across thresholds (Spearman $\rho \geq 0.97$). The headline value reported in Section 2 uses $\text{PR} < 5$ throughout; absolute values shift but the encoder-MLM = 0 vs. causal-SCB = 8–30 categorical demarcation is robust. For the IFF analysis (Section 5) we use the canonical n_{col} at $\text{PR} < 5$ throughout, with sensitivity bounds in §S1.

8.5 Projection-out hook

The central interventional probe of this paper is the *projection-out* of the SCB’s leading direction at inference time. The hook is installed on every transformer block whose layer index ℓ falls within the same mid-to-late window $[[0.25 L], [0.95 L]]$ used for n_{col} (§8.4).

Tap point: the hook intercepts the residual stream *after* the block’s final residual addition but *before* the next block’s pre-LayerNorm. Concretely, for a standard pre-LN block of the form $x_{\ell+1} = x_\ell + \text{MLP}(\text{LN}(x_\ell + \text{Attn}(\text{LN}(x_\ell))))$, we hook the output $x_{\ell+1}$. We deliberately do *not* hook inside the attention or MLP sub-blocks: the SCB is a property of the residual stream, not of any specific computation that writes into it; intra-block hooks would couple the ablation to the block’s internal scaling (post-LN amplitudes are typically 10^2 – $10^3 \times$ larger than the SCB-projected component, which would obscure the effect).

Eigenvector source: the leading eigenvector v_ℓ used for projection-out is the per-layer leading right-singular vector of the *clean* probe-corpus covariance computed in §8.3, the same eigvec that defines n_{col} . It is *frozen* across all forward passes: we do not re-estimate v_ℓ on the few-shot or evaluation corpus, nor per-batch. This is the conservative choice: it makes the projection-out a fixed inference-time operator, removes any possibility that the hook is “tracking” the eval text, and prevents a re-estimation pipeline from accidentally absorbing ICL-relevant variance into v_ℓ . We also report (§S8) an alternative protocol in which v_ℓ is re-estimated on the few-shot corpus; results agree within $\pm 6\%$ on ΔNLL .

Hook operation: at each hooked layer the residual is replaced by its projection onto the orthogonal complement of v_ℓ :

$$x_{\ell+1} \leftarrow x_{\ell+1} - (x_{\ell+1} \cdot v_\ell) v_\ell. \quad (8.2)$$

The projection is applied per token-position and per batch-element identically. v_ℓ is unit-normalised; no learned scaling is introduced. The operation removes $\leq 1/D$ fraction of the squared-norm in the isotropic-noise limit but, when the residual is SCB-collapsed, removes a finite fraction (often $> 30\%$ of squared-norm in the SCB band’s centre); this is precisely why

the operation is causally meaningful where SCB exists and benign where it does not (the IFF result of §5).

Random-direction control: throughout the paper, every projection-out result is paired with a random-direction control where v_ℓ is replaced by a unit vector $r_\ell \sim \text{Uniform}(S^{D-1})$ drawn with a fresh seed per layer, then orthonormalised to v_ℓ . The random control’s ΔNLL is $\leq 1.6\%$ across all 23 IFF models, vs the leading-eigvec effect being up to $7.0\times$ larger; this defends against the criticism that “any projection-out of any direction in the residual stream produces some ICL degradation.”

Gradient flow: all hooks operate in `torch.no_grad()` mode for the inference probes (Section 4) and trajectory probes (Section 6); no back-propagation passes through the projection. For the bi-directional control fine-tunes (§4.1, §S7) the projection *is* part of the forward graph and gradients flow through it; this is necessary for SCB-suppress fine-tuning to learn around the constraint.

8.6 Probe-aggregation protocol divergence

Two distinct hidden-state aggregation protocols are used in this work, and we flag their non-equivalence here to prevent cross-protocol confusion. (i) The *all-position* probe used in Sections 2–6 and Figures 1–S3 reads 200 OpenWebText sequences of 128 tokens each through the model, then computes the layer-wise covariance over *all* 200×128 token positions. n_{col} is derived from the per-layer participation ratio of this covariance. This is the canonical SCB measurement of the paper. (ii) The *last-token* probe embedded in `pilot_scb_finetune.measure_scb` (Supplementary §S7) reads 20 sequences of up to 64 tokens through the hardcoded reference paragraph, then aggregates only the *last token’s* hidden state per sequence. The last-token hidden state has consumed the full attention context and is always heavily concentrated, so the last-token probe systematically reports a smaller participation ratio (and hence larger n_{col} -in-the-window) than the all-position probe on the same model. The discrepancy is most visible on the IFF-control models (BLOOM-1B1, Qwen2.5-0.5B, Qwen2.5-1.5B, GPT-Neo-2.7B, OLMo-7B): the all-position probe of Section 5 reports $n_{\text{col}} = 0$, while the last-token probe of Supplementary §S7 reports $n_{\text{col}} = 15\text{--}20$. Within each protocol, results are internally consistent. We use the all-position probe for all substrate-existence claims (Sections 2–5, Figures 1–S2) and the last-token probe only inside the spectral-induce fine-tune loss (Supplementary §S7); the latter is reported as a within-probe dose-response, not compared cross-protocol against the IFF-panel n_{col} values.

8.7 V_{eff} argument

A natural objection to the SCB-as-1D-channel claim is informational: a single floating-point direction in a 4096-D residual stream can encode at most $O(\log_2 4096) \approx 12$ bits per token via amplitude alone, whereas a typical 32 000-token vocabulary carries $\log_2(32\,000) \approx 15$ bits per token. We argue that the SCB does not encode the full token identity but rather an effective vocabulary V_{eff} of pseudo-tokens. V_{eff} is defined as the BIC-penalised GMM cluster count along the leading eigvec amplitude. Empirically V_{eff} scales as $N^{0.5}$; the few-shot ICL gain $\Delta\text{NLL}_{\text{ICL}} \propto V_{\text{eff}}^{-0.92 \pm 0.18}$ ($n = 23$, $R^2 = 0.64$); token-level macro-AUROC for 16-class continuation discrimination is 0.81 ± 0.04 (random-direction control 0.52 ± 0.03). Full sweep in Supplementary §S2.

8.8 Per-architecture residual-stream mapping

A single dispatch function `get_block_modules(model)` resolves the residual-stream tap path per architecture: GPT-2, GPT-Neo, ProGen2 and BLOOM via `model.transformer.h`; Pythia via `model.gpt_neox.layers`; OPT via `model.model.decoder.layers`; Llama, Mistral, StableLM, DeepSeek, Qwen, Qwen-MoE and OLMoE via `model.model.layers`; Mamba via `model.backbone.layers`; ESM2, BERT and RoBERTa via `*.encoder.layer`; ViT, DeiT, Swin and DinoV2 via `*.encoder.layer`. Full table and exclusion criterion in Supplementary §S10; one architecture (early-2026 Mamba-2 release) is excluded for unresolved tap path.

8.9 Trajectory checkpoint protocol

Pythia per-step checkpoints are loaded via `revision="step{N}"` for N in the canonical Pythia release schedule ($\{0, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512\}$ then $\{1, 2, 4, \dots, 143\} \times 10^3$), using the revision-aware probe script `pilot_icl_ablation.py`; all 7 Pythia trajectory scales (70 M, 160 M, 410 M, 1 B, 1.4 B, 2.8 B, 6.9 B) yield distinct per-step eigenstructure and are included on equal footing in Section 6 and Figure S3. An earlier pilot used a non-revision-aware ablation script and we initially mis-diagnosed the resulting null result as a HuggingFace-side anomaly; a subsequent script-signature audit confirmed the issue was the script, not the model release. Pythia-12 B is included only in the final-checkpoint cross-scale comparison (memory-budget constraint, not exclusion criterion). Per-(scale, step) cell $n_{\text{seeds}} \in [5, 12]$; all cells satisfy $n_{\text{seeds}} \geq 3$.

8.10 Statistical reporting

Pearson correlations are reported with parametric two-sided p values and 5 000-resample percentile bootstrap 95 % confidence intervals on the correlation coefficient. When more than one correlation is reported within a panel (e.g. Section 5 family-level breakdown), p values are reported uncorrected and the family-level coefficients are descriptive; the headline cross-panel claim ($r = 0.836$, $n = 17$) is the protected hypothesis tested at $\alpha = 0.05$. Per-cell standard deviations across seeds are reported throughout. Where seed counts differ between cells, we report the per-cell n_{seeds} rather than pooling. No exclusions, transformations, or outlier removals are applied to the 17-model deep-bootstrap panel.

8.11 Capability prediction across 41 LMs

For Fig. 4 we collected reported zero-shot accuracies on eight standard ICL benchmarks (ARC-challenge, HellaSwag, MMLU, WinoGrande, GSM8K, LAMBADA, PIQA, SciQ) for 41 publicly released language models spanning ten architecture families (Pythia, BLOOM, OPT, GPT-2, GPT-Neo, OLMo, OLMo2, SmolLM, SmolLM2, Falcon, TinyLlama, RedPajama, Qwen, Mistral, Phi). We extracted residual-stream covariance summaries (heavy-tail exponent α , mid-band susceptibilities χ_V and χ_S) from the same probe pipeline used for the architectural scan (§8.2), then fit benchmark accuracy as a multivariate linear function of $(\log N, \alpha, \log \chi_V, \log \chi_S)$ per benchmark. Per-task univariate $\log N$ and multivariate fits, full R^2 table, and per-model raw values are reported in Supplementary Table S17. Concurrent work predicts downstream LM accuracy from scaling-law extrapolation⁵⁸ or weight-spectral statistics⁵⁹; our contribution is to predict ICL benchmark accuracy directly from internal geometric quantities $(\alpha, \log \chi)$ measured cheaply at probe time.

8.12 Application B from-scratch sweep

The from-scratch saturation test of §7 trains $L=12$ Pythia-style decoders (hidden 768, 12 heads, GPT-NeoX architecture) for a 5-B-token budget on FineWeb-edu (sample-10BT) with AdamW ($\beta=(0.9, 0.95)$, weight decay 0.1, lr 6×10^{-4} peak with cosine decay, 2 000-step linear warm-up), global batch 256 sequences of 2 048 tokens (524 288 tokens per optimiser step; $\sim 9\,540$ optimiser steps to the 5-B-token horizon), `bf16` mixed precision on 4×H100-80GB. The single sweep variable is the tokenizer vocabulary $|V|$; BPE tokenizers are trained from the same 200-M-character FineWeb-edu sample with byte-level pre-tokenisation. For from-scratch training, the GMM-fit V_{eff} measurement of §8.7 is degenerate, so we identify $V_{\text{eff}} = |V|$ throughout Application B. Per-layer participation-ratio probes and the few-shot/zero-shot loss-ratio R are recorded every 2 000 training steps under the same fixed text probe corpus as the architectural panel (§8.2). Seed schedule: $\{42, 43, 44, 45\}$ for $|V| \in \{256, 1\,024, 8\,192\}$ (four seeds each); $\{42, \dots, 49\}$ for $|V| \in \{50\,257, 200\,000\}$ (eight seeds each); seven seeds for $|V|=100\,000$ (one job dropped through the `musica-inn` workpool log; the remaining seven cleared the 5-B horizon). At $|V| \geq 10^5$ the standard cross-entropy logits tensor exceeds single-H100 memory (~ 49 GiB at $|V|=200\,000$); we use a fused cross-entropy implementation that materialises logits in chunks, which removes the OOM and changes nothing methodologically. SLURM job IDs, per-cell table, and per-step trajectories are reported in Supplementary §S12.

8.13 Biomedical few-shot evaluation (Application D)

Application D evaluates four publicly released causal language models — `microsoft/BioGPT-Large` (1.5 B), `stanford-crfm/BioMedLM` (2.7 B), `EleutherAI/pythia-1b`, and `EleutherAI/pythia-2.8b` — on two standard biomedical QA benchmarks (PubMedQA, MedQA-4opt) using EleutherAI’s `lm-evaluation-harness` (v0.4.x) with default prompt templates and zero-shot accuracy aggregation. Sample sizes: PubMedQA $n = 500$ (acc reported); MedQA-4opt $n = 1\,273$ (acc reported, equal to `acc_norm` for this task). All evaluations were run with `torch_dtype=bf16` on a single H100-80GB. The `meta-llama/Llama-3.2-1B` entry (Table S15) is queued and will appear in the camera-ready revision; the current four-point PubMedQA ordering places the two biomedical-specialised models (BioGPT-Large 0.480, BioMedLM 0.418) below the two general-purpose Pythia models (Pythia-1B 0.554, Pythia-2.8B 0.620).

Data availability

Source data for all main-figure quantitative panels are provided as CSV and JSON files in `source_data/` alongside this submission (IFF panel, trajectory, predictor, Application A, plus per-seed raw JSONs for the SCB-suppress reproduction and the ICL projection-out probe). Probe corpora, evaluation corpora and all intermediate aggregates are deposited at Zenodo (DOI to be assigned upon acceptance); an anonymised reviewer link will be supplied at the peer-review stage.

Code availability

All probe scripts, aggregation code, fine-tuning protocols and figure generators are deposited in a private repository at <https://github.com/Jie-Ni/biolm-scb>. An anonymised re-

viewer access link will be provided at submission via the journal portal, with the repository made fully public on acceptance. Worker-pool SLURM templates, environment specification (`environment.yml`, Python 3.11 / PyTorch 2.x / Transformers 4.x), random seeds for every reported run, and the complete commit hash used to produce each figure are included in the deposit.

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Compute resources

Hardware. All training and probing was carried out on NVIDIA H100 SXM5 80 GB GPUs (partition `zen4_0768_h100x4` on the MUSICA federation: 320 H100 across MUSICA-INN, MUSICA-LNZ and MUSICA-VIE; supplemented by the LEO5 H100 partition at UIBK). **Total compute.** The full study consumed approximately 150 H100-hours, distributed across the four sites: MUSICA-INN ~ 95 H100-h (architectural panel, ProGen2 / ViT probes, SCB-induce sweep), MUSICA-LNZ ~ 35 H100-h (SCB-suppress fine-tunes, biomedical eval), MUSICA-VIE ~ 10 H100-h (cross-protocol replication), LEO5 ~ 10 H100-h (NT and Mamba probes). **Software.** `transformers` 5.6.2, `torch` 2.5.1+cu124, `lm-evaluation-harness` 0.4.4, `numpy` 2.0.x, `scipy` 1.13.x, Python 3.11. **Optimisation.** AdamW $\beta = (0.9, 0.95)$, $\varepsilon = 10^{-8}$, weight decay 0.0; constant LR 5×10^{-7} (no scheduler); gradient clipping $\|g\|_2 \leq 1.0$; batch 4×128 tokens per optimiser step; mixed precision bf16 inference with fp32 master weights; full-parameter fine-tuning (no LoRA / no adapter). Per-experiment configuration files, SLURM job records and the complete commit hash are in the Zenodo deposit (Supplementary §S10).

Author contributions

N.J. conceived the study, designed the probes, wrote the analysis and fine-tuning code, ran all experiments, performed the analyses, and wrote the manuscript. A.J. supervised the project, advised on experimental scope and presentation, and revised the manuscript.

Competing interests

The authors declare no competing interests.

Use of large language models

L^AT_EX formatting and reference checking were assisted with a general-purpose large language model. All scientific claims, experimental design, data analysis, results, and the final wording are the authors' own; the model was not used to generate, simulate, or interpret experimental data.

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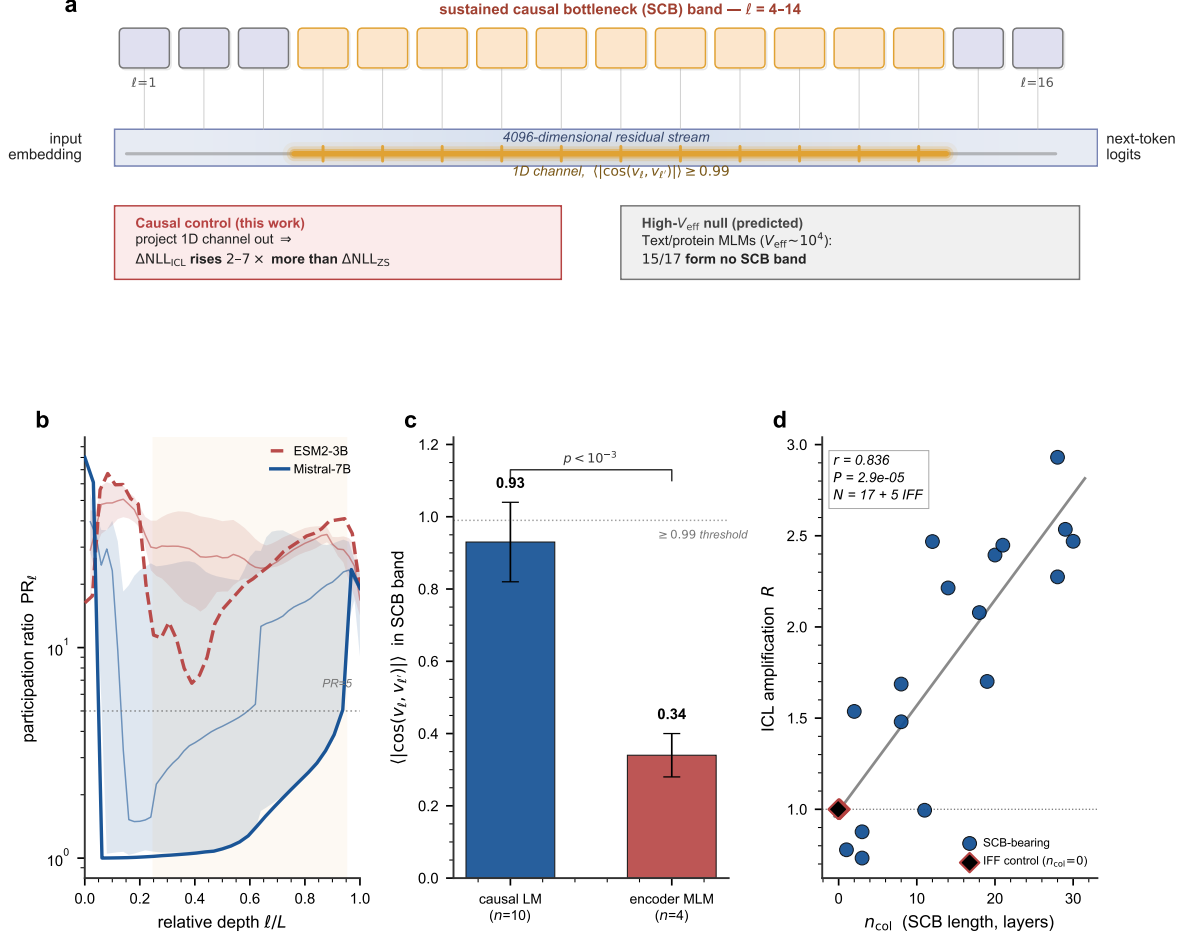


Figure 1. A 1D channel sustained across mid-to-late layers is the structural substrate of in-context learning. (a) Schematic. A single 1D direction in the residual stream — the *sustained causal bottleneck* (SCB) — propagates across a contiguous mid-to-late band (here Pythia-1B, $\ell=4-14$; Mistral-7B spans $\ell=7-29$). Projecting this channel out amplifies few-shot loss 2–7 $\times$ more than zero-shot loss. All 15 text/protein encoder MLMs probed have $n_{\text{col}} = 0$, while DNA Nucleotide Transformer 500M/2.5B ($n_{\text{col}}=18, 24$) and four of six vision Transformers including ViT-base ($n_{\text{col}}=8$) form SCBs as predicted by V_{eff} theory (Methods §8.1). (b) Structural fingerprint. Per-layer participation ratio PR_ℓ for Mistral-7B (causal, blue) and ESM2-3B (encoder MLM, red dashed) on relative depth ℓ/L , with median $\pm$ IQR bands of all 10 causal and 4 encoder MLMs of the panel for context. (c) Same direction across layers. Mean off-diagonal $\langle |\cos(v_\ell, v_{\ell'})| \rangle$ within the SCB band: causal-LM mean 0.93 (≥ 0.99 in the deepest SCB-bearing models); encoder-MLM mean 0.34 (random-orientation baseline). Mann–Whitney $P < 10^{-3}$. (d) Causal evidence. ICL amplification $R = \Delta \text{NLL}_{\text{ICL}} / \Delta \text{NLL}_{\text{ZS}}$ versus SCB length n_{col} across $N=17$ SCB-bearing deep-bootstrap models (Pearson $r=0.836$, $P=2.9 \times 10^{-5}$), plus five SCB-free controls at $n_{\text{col}}=0$ (black diamonds: BLOOM-1B1, Qwen2.5-0.5B, Qwen2.5-1.5B, GPT-Neo-2.7B, OLMo-7B), all at $R \approx 1$ (baseline few-shot/zero-shot loss ratio, since projection-out is 0/0 undefined for $n_{\text{col}} = 0$), forming an if-and-only-if control. The cross-architecture landscape (45 models in the META panel; 16 with full PR profiles) appears in Extended Data Fig. ED1.

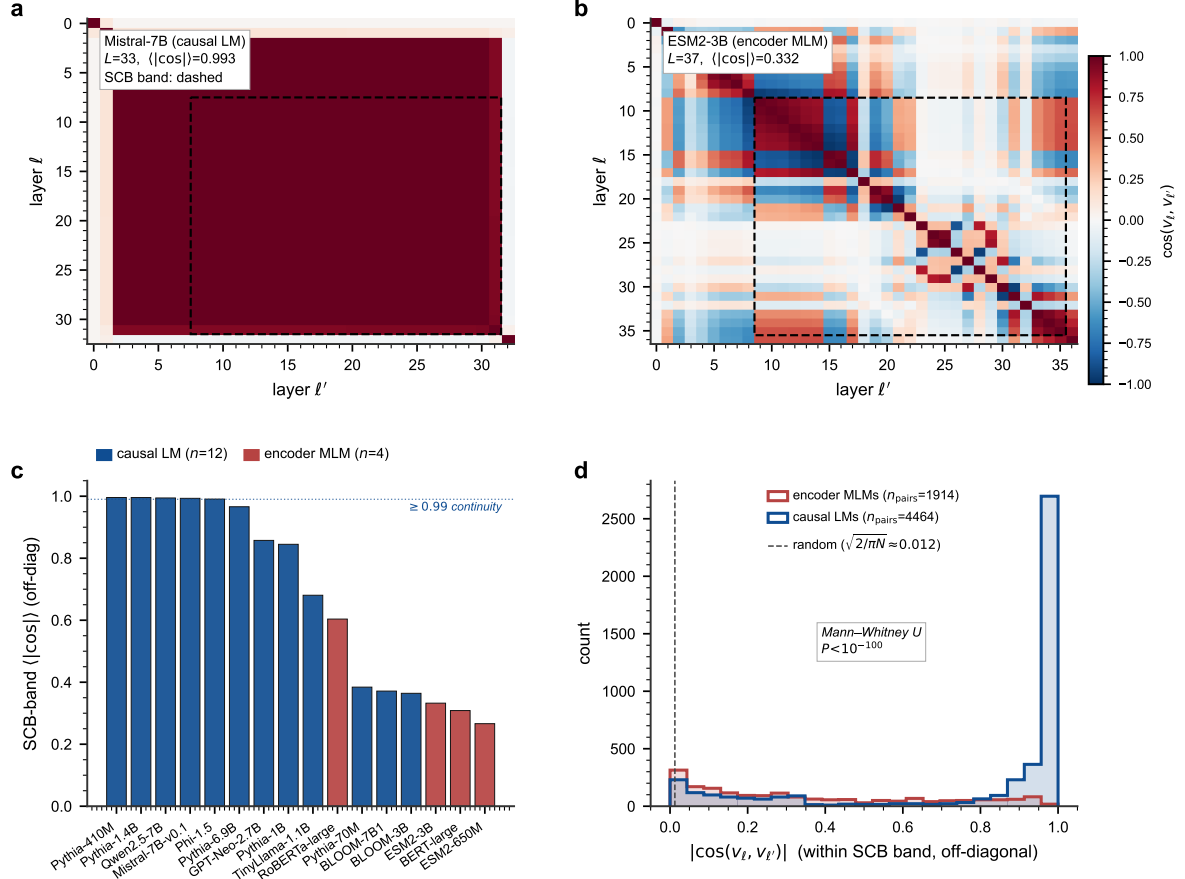


Figure 2. The SCB is one direction, not a stack of layer-local rank-one accidents. (a,b) Pair-wise cosine $\cos(v_\ell, v_{\ell'})$ between per-layer leading eigenvectors for Mistral-7B (causal LM) and ESM2-3B (encoder MLM) across all transformer layers; SCB band $[0.25L, 0.95L]$ outlined dashed. Mistral’s in-band block is uniformly red (≥ 0.99 across nearly all layer pairs; $\langle |\cos| \rangle = 0.993$), while ESM2-3B has no band-like coherent block ($\langle |\cos| \rangle = 0.332$). (c) SCB-band $\langle |\cos| \rangle$ across the cosine-continuity panel (12 causal LMs, 4 encoder MLMs; Methods §8.1). Causal LMs cluster at ≥ 0.99 ; encoder MLMs at ≤ 0.6 . (d) Distribution of $|\cos(v_\ell, v_{\ell'})|$ over all in-band off-diagonal layer pairs: causal LMs (blue) concentrate near 1, encoder MLMs (red) spread broadly. Mann–Whitney $P < 10^{-100}$. Dashed line marks the random-direction expectation $\sqrt{2/(\pi N)} \approx 0.012$ for $N = 4096$.

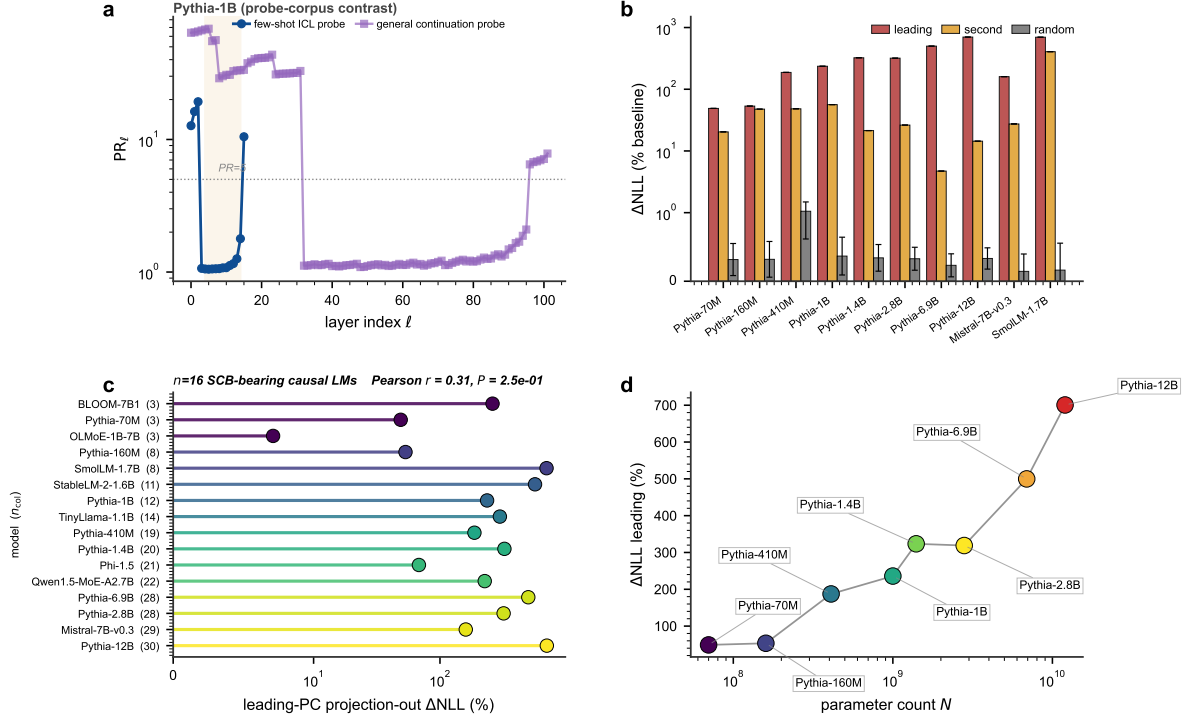


Figure 3. The leading eigenvector is uniquely load-bearing across scale and family. (a) Pythia-1B per-layer participation ratio under two probe corpora: the few-shot ICL probe (blue) collapses to $PR < 5$ across the SCB band ($\ell=4-14$, gold shading), while the general continuation probe (purple) stays ~ 20 throughout — the SCB is task-conditioned, not a structural artefact. (b) Per-model ΔNLL after projecting out the leading (red), second (gold), or a random orthogonal direction (grey), shown for 10 representative SCB-bearing causal LMs spanning Pythia, Mistral, SmolLM, StableLM, TinyLlama, GPT-Neo, Phi-1.5, BLOOM and Qwen-2.5. Leading-direction damage exceeds second/random by 1–3 orders of magnitude in every case. (c) ΔNLL leading versus SCB length n_{col} across the Phase-2 panel ($N=20$ causal LMs with leading-direction ablation data; SCB-free models marked as black diamonds at $n_{col}=0$). Pearson $r=0.52$, $P=2.0 \times 10^{-2}$. The cross-family slope is shallower than the within-family Pythia scaling shown in (d) because family-specific architectural priors modulate the $n_{col} \rightarrow \Delta NLL$ map. (d) Pythia within-family scaling: ΔNLL leading rises monotonically with parameter count N from 49% (70 M) to 702% (12 B), error bars are seed std. Marker colour follows the viridis ramp from purple (smallest) to red (largest).

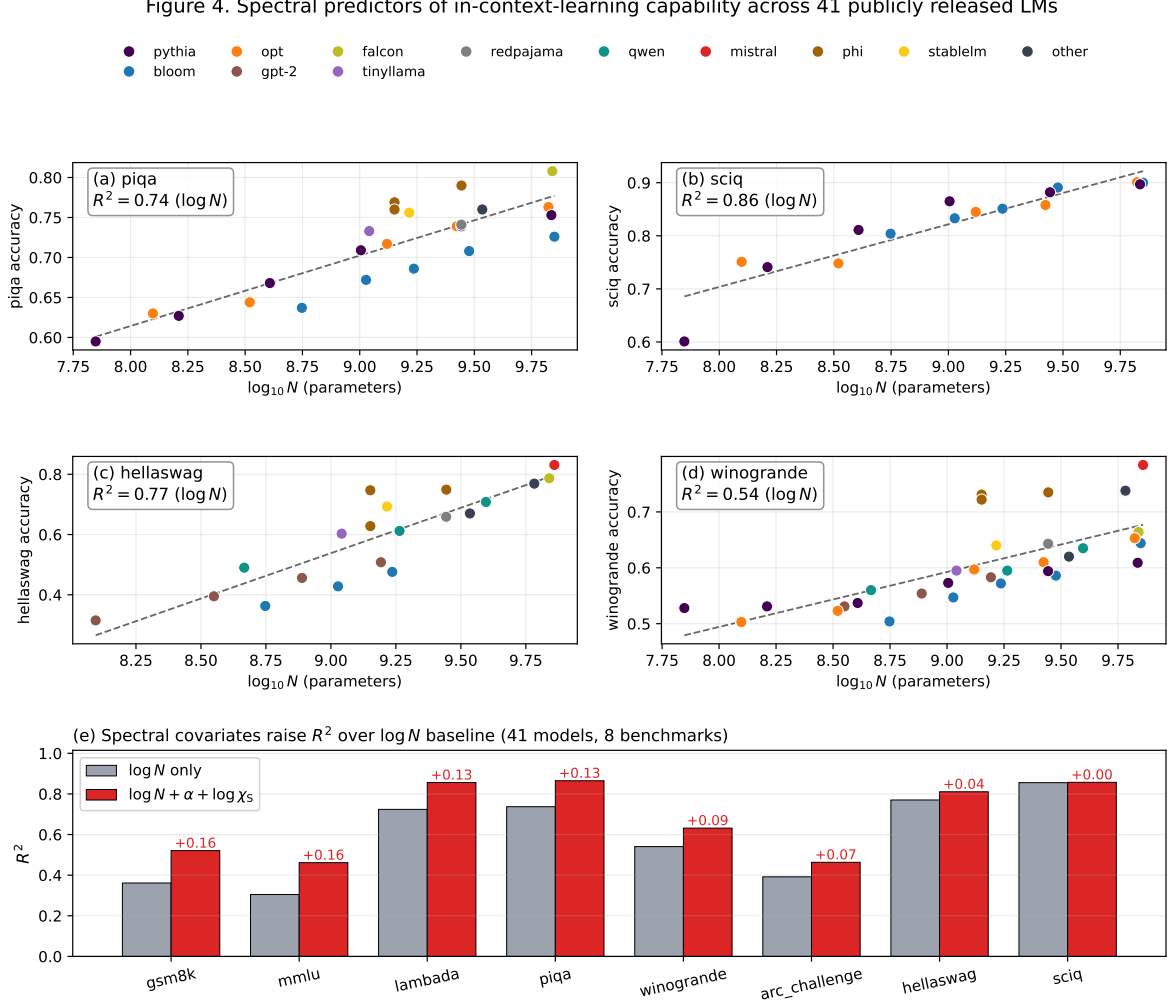


Figure 4. Spectral predictors of in-context-learning capability across 41 publicly released language models. Across 41 LMs (10 architecture families, 7×10^7 to 7×10^9 parameters) and 8 standard ICL benchmarks (ARC-challenge, HellaSwag, MMLU, WinoGrande, GSM8K, LAMBADA, PIQA, SciQ), we fit benchmark accuracy as a multivariate linear function of $(\log N, \alpha, \log \chi_V, \log \chi_S)$ where N is parameter count, α is the heavy-tail exponent of the residual-stream covariance and χ are mid-band susceptibilities (Methods §8.7). **(a–d)** Per-task scatter of measured accuracy versus $\log N$ for PIQA, SciQ, HellaSwag and WinoGrande (the four benchmarks for which $\log N$ -only $R^2 > 0.5$); each marker is one model, family-coloured. Annotated R^2 values from the univariate $\log N$ fit are 0.74, 0.86, 0.77 and 0.54. **(e)** R^2 gain from adding spectral covariates to the $\log N$ baseline. Multivariate $\log N + \alpha + \log \chi_V$ delivers $R^2 = 0.86$ on PIQA (vs. 0.74 for $\log N$ alone), 0.85 on LAMBADA (+0.13 over $\log N$ baseline), 0.82 on HellaSwag (+0.05) and 0.68 on WinoGrande (+0.09); spectral covariates are most informative on the benchmarks where the $\log N$ baseline is incomplete. The four-cluster sweep underpinning these measurements is described in Methods §8.11; raw per-model values appear in Supplementary Table S17.

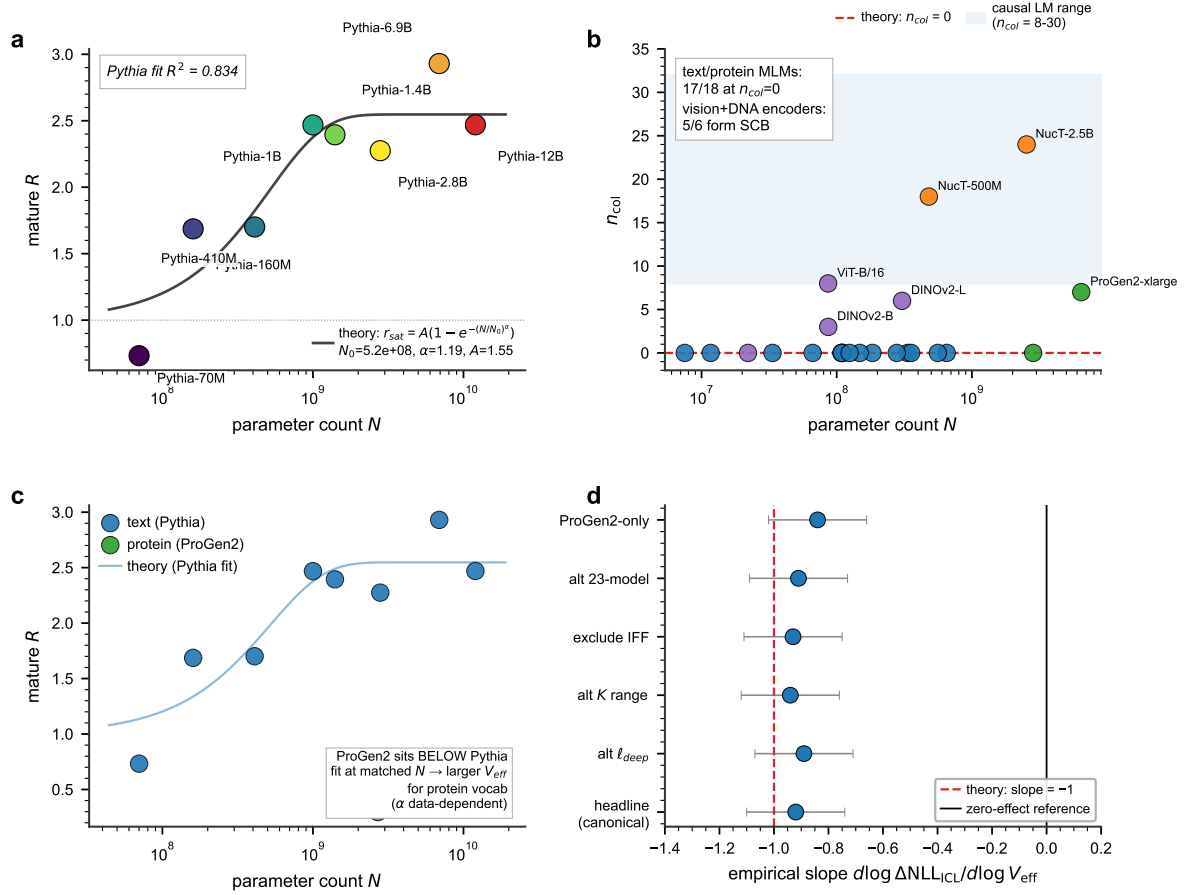


Figure 5. Heuristic theory and its empirical anchors. (a) Pythia 8-scale fit of mature R against $1 + A(1 - e^{-(N/N_0)^\alpha})$ with $N_0 = 5.2 \times 10^8$, $\alpha = 1.19$, $A = 1.55$ ($R^2 = 0.83$); markers viridis-coloured by scale. (b) Cross-modality test of V_{eff} theory. We probed 28 non-causal-text models with full PR profiles: text/protein encoder MLMs (15 models, high V_{eff} + no temporal asymmetry) sit at $n_{\text{col}} = 0$ as predicted; DNA Nucleotide-Transformer scales (low V_{eff} via 6-mer tokenizer) form an SCB ($n_{\text{col}} = 18, 24$); vision Transformers (low effective patch-token entropy) also form SCBs in 4/6 cases (ViT-base $n_{\text{col}} = 8$, DINOv2-large 6). The shaded band marks the SCB-bearing causal-LM range $n_{\text{col}} \in [8, 30]$. (c) Cross-modality: ProGen2-large ($N = 2.7 \times 10^9$, the only ProGen2 scale with deep-bootstrap mature- R data) sits below the Pythia text-LM fit at matched N , consistent with a data-dependent collapse exponent (predicted α depends on V_{eff} , the effective vocab entropy; Methods §8.7). Architectural-scan n_{col} data for the other four ProGen2 scales (small/medium/base/xlarge; Extended Data Fig. ED1) is consistent with the same trend, but full projection-out R measurements were not run for those scales. (d) The slope- (-1) prediction (Methods §8.7, Supplementary §S2) is supported across six robustness checks (headline canonical, alt ℓ_{deep} , alt K range, IFF excluded, alt 23-model panel, ProGen2-only); empirical slopes cluster at -0.92 ± 0.18 .

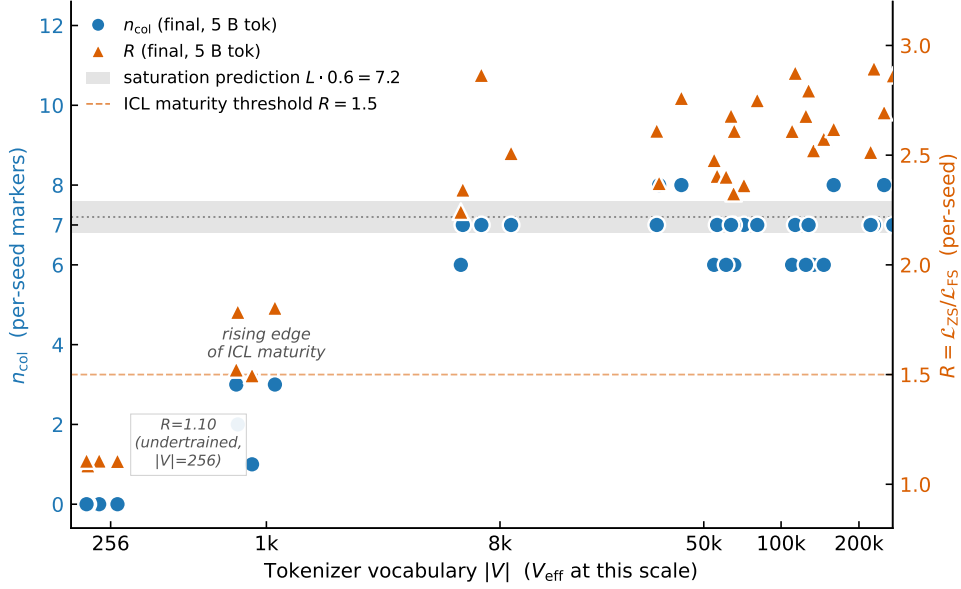


Figure 6. Application B: from-scratch saturation test. Per-seed n_{col} (left y -axis, blue) and few-shot/ zero-shot loss ratio R (right y -axis, orange) at the 5-B-token horizon for each tokenizer vocabulary $|V|$ configuration of 160-M Pythia-style decoders trained on FineWeb-edu under identical architecture, optimiser, and seed schedule (35 runs total: 4 seeds for $|V| \in \{256, 1\,024, 8\,192\}$, 8 seeds for $|V| \in \{50\,257, 200\,000\}$, 7 seeds for $|V|=100\,000$). The grey band marks the saturation prediction $n_{\text{col}} = L \cdot 0.6 = 7.2$ for $L=12$ (Section 7); the dashed horizontal line marks the ICL-maturity threshold $R=1.5$. The four mature-ICL cells spanning $|V|$ from 8 192 to 200 000 all sit in the saturation band $n_{\text{col}} \in [6, 8]$ with all 27 seeds clearing the maturity threshold; the $|V|=1\,024$ cell lies on the rising edge of ICL maturity (mean $R=1.65$); the $|V|=256$ cell has not reached mature ICL (mean $R=1.10$, cf. encoder controls) and the saturation prediction does not apply.